

Introduction

Sports analytics is having a growing impact on the decisions being made in the NFL and football in general, changing how organizations assess player worth, create offensive strategies, and assign roster slots. As statistical techniques have evolved, NFL offenses have shifted toward pass-heavy strategies and smaller personnel groups, frequently disregarding conventional positions like fullback (Katbi & Kleman; Hughes et al., 2015). As a result, fullback use fell significantly in the early 2010s. Thankfully, over the last few seasons, there has been a considerable increase in fullback usage, with one-third of NFL teams using fullbacks in their offensive schemes. This move presents a crucial analytical question, are offenses making a strategic mistake by not utilizing a fullback in their offense, and how can analytics support or not support the value of the fullback?

Evolution of Offenses

An enormous collection of sports analytics research identifies passing effectiveness as the key factor of performance in the NFL. Katbi and Kleman looked at league-wide data from 1990 to 2008 and discovered that being efficient in the pass has a much better correlation with wins than being efficient in the run. Although rushing efficiency remained highly significant, its relative impact on winning games was significantly lower. This research influenced a league-wide ideological change toward spread offenses and personnel packages designed for passing success. Since analytical models prioritized passing efficiency, organizations adapted by focusing on wide receivers, quarterbacks, and pass-protecting linemen while cutting back on typical running-friendly positions like fullbacks (Katbi & Kleman). While this work clearly illustrates the analytical reason for the fall of the fullback, it focuses mostly on offensive efficiency as an outcome rather than how efficiency is achieved within certain personnel formations.

Run Game Efficiency

Despite the prevalence of passing productivity in current data analysis, rushing efficiency is still a significant contribution to offensive performance. Many studies evaluate rushing performance using collective metrics like yards per attempt or total rushing production (Katbi & Kleman). However, these measurements fail to account for the underlying factors that result in effective rushing results, such as blocking tactics, advantage, and the existence of lead blockers. Fullbacks are often assigned to bigger personnel groups, like twenty-one personnel (two backs, one tight end), and are rarely examined independently. As a result, the research offers little insight into how fullbacks might improve rushing effectiveness through lead blocking, defensive manipulation, or increased yards before contact. This drawback shows that, although rushing efficiency seems to have been less valuable than passing efficiency in collective models, measurement limits might underestimate the unique roles that enable a successful running game.

Strategic Play Calling

Current sports analytics research has viewed football as a statistical decision-making problem, with an emphasis on selecting the best play based on game context and projected outcomes. Studies on play-calling tactics emphasize the importance of situational awareness, such as down, distance, and field position, in determining offensive performance. Yet many play-calling models are based on set personnel packages and do not take into consideration how personnel influence a variety of potential strategic choices. The inclusion of a fullback may increase offensive flexibility by allowing for a variety of play calls, such as downhill runs, play-action passes, improved pass protection, and matchup-based route ideas (depending on the athleticism of the fullback). While current studies assess the effectiveness of specific play calls, it rarely examines how various player roles impact decision-making capabilities inside offensive schemes.

Value of positions

Hughes et al. (2015) created the Wins Above Replacement (WAR) methodology, which has had a significant impact on NFL position evaluation. Using injuries and suspensions as random variables, the authors calculate the value of NFL positions based on changes in team performance. According to Hughes et al. (2015), the combination of the tight end/fullback position group is among the most valued offensive positions, along with quarterbacks, wide receivers, and offensive tackles. This finding is especially noteworthy given the decreased use of fullbacks during the research period. The findings indicate that certain situations may provide contextual value that is difficult to measure using typical statistics. However, because tight ends and fullbacks are lumped together, the study does not separate fullbacks' independent contributions, which limits findings on their own analytical value.

Methods and limitations in Sport Analytics

A larger body of sports analytics research highlights the necessity of assessing player contribution beyond the box score stats. Fundamental work in football analytics emphasizes the importance of process-driven evaluation methodologies that account for situational inputs rather than depending exclusively on visible outputs. These techniques imply that positions having indirect influence, such as blocks or hybrid offensive roles, may be routinely devalued when evaluation systems prioritize simple to measure outcomes. Furthermore, player analysis relating to the NFL Combine shows that athletic testing metrics do not necessarily directly transfer to on-field performance or positional worth. This lends credibility to the theory that certain positions, such as fullbacks, are devalued due to biases in grading methodologies rather than a lack of functional significance.

The Gap

Even though previous sports analytics work provides suggestive evidence supporting the importance of passing effectiveness, situational decision-making, and positional value in the NFL, little thought has been paid to identifying the fullback position's distinct statistical impact. Prior research has frequently grouped fullbacks with tight ends (Hughes et al., 2015), evaluated rushing effectiveness without investigating its underlying mechanisms (Katbi & Kleman), or modeled play-calling decisions without accounting for personnel-driven versatility. As a result, there is still a lack of awareness of fullbacks' relative impact on offensive success in both the rushing and passing game. This disparity is especially significant given the recent rise of fullback usage in current NFL schemes, and it prompts more analytical examination into whether offenses may have purposefully undervalued the position.

Methodology

The data set that was used in the research came from `nflreadr`, which is extracted from NFL play by play data. The seasons used were from 2020-2024, since this is where fullbacks had play by play data imputed. The play by play data includes many factors like down, play type, outcome, play type, and many advanced features as well. A secondary data set came from `nflreadr` as well, but this was the participation data where I was able to see offensive and defensive personnel in order to obtain when a fullback was in the game. Using these datasets, I was able to engineer new features to fit what I wanted in my regression models. The first variable created was `has_fullback`, it is binary and was obtained by using the offensive personnel string “ 1 FB”. This variable shows whether a fullback was in the game during the play. So every play that had that string was deemed `has_fullback`. The next variable created was `short_yardage_success`. If this equals 1 meaning the play gained one yard, it was considered a success, since in these situations all you want to get is one yard. This variable was also binary, 1 being successful, and 0 not being successful meaning they didn't gain a yard. The other variable created was `goal_line_success`. Like the short yardage and `has_fullback`, variable it is also binary. 1 is considered a success being if a touchdown was scored, and 0 being not successful, meaning a touchdown was not scored. The data was then divided into three situations, short yardage, goal line, and base down. In short yardage it was considered third or fourth down, with two or fewer yards to go. Goal line was considered a team being on their opponents five yard line and in. Then base downs were considered first down and ten, and second down and between five yards and ten yards. Only the plays in these situations were used in the analysis. Using Pearson Correlation, features used in all of the models were determined by this. This is calculated by :

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2} \sqrt{\sum(y_i - \bar{y})^2}}$$

x_i y_i , represents the observations of two variables and $\bar{x}\bar{y}$, show the mean of each variable.

The range is from -1 to 1, which are the strongest correlations either negative or positive. For short yardage situations the features used in the correlation, with respect to short_yardage_success, were, has_fullback, down, first_down, first_down_rush, ydstogo, yardline_100, score_differential, defenders_in_box, shotgun, no_huddle, qtr, wp, wpa, ep, td_prob, fg_prob, xpass, posteam_score, defteam_score, drive_play_count, and drive_first_downs. The same features were used in goal line situations with respect to goal_line_success, this also holds true for base down situations with respect to EPA(expected points added).

Next were the actual regression models created for the analysis. The first model was the short yardage models, which contains three parts, part one being the short yardage grouped together, the second being just run plays, and the third being just pass plays in this situation. The features used in these models differed, but starting with short_yardage_all, the features used were has_fullback, down, ydstogo, yardline_100, defenders_in_box, first_down_rush, first_down, wpa, drive_first_downs, drive_play_count, xpass. In the short yardage rushing all the features remained the same. Where the features changed a slight bit were in short yardage passing plays, and the feature that was taken out was first_down_rush. These performance models used logistic regression since the outcome was short_yardage_success, which is binary. As stated before it estimates whether a play is successful or not using the logit function. The abbreviated models can be shown below:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1(has_fullback) + \beta_2(down) + \beta_3(ydstogo) + \beta_4(yardline_100) \dots + \beta_{11}(xpass)$$

The goal line situations also used logistic regression, since goal_line_success is a binary outcome as well. The features used in this model included has_fullback, wpa, epa, yardline_100, ydstogo, first_down_rush, first_down, drive_first_downs, drive_play_count, drive_inside20, success, series_success, no_score_prob. The same features for goal line rushing and passing followed the same features as the all model. This can be represented as this:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1(has_fullback) + \beta_2(wpa) + \beta_3(epa) + \beta_4(yardline_100) \dots + \beta_{11}(no_score_prob)$$

The base down model had the outcome variable, EPA(expected points added). This is a continuous variable so linear regression was applied to this model. The features used in this model were has_fullback, wpa, ep, ydstogo, first_down_rush, first_down, drive_first_downs, drive_play_count, xpass, down, defenders_in_box, shotgun, rush_attempt, air_epa, yac_epa, wp, cp, cope, pass_oe, defense_man_zone_type, defense_coverage_type. The base rush has everything expect for air_epa, yac_epa, cp, cope, pass_oe, defense_man_zone_type, defense_coverage_type, run_location was also added to this. Passing plays had everything in the base model expect the rushing features. This is represented by:

$$EPA = \beta_0 + \beta_1(has_fullback) + \beta_2(wpa) + \beta_3(first_down) + \beta_4(first_down_rush) \dots + \beta_{11}(defense_coverage_type)$$

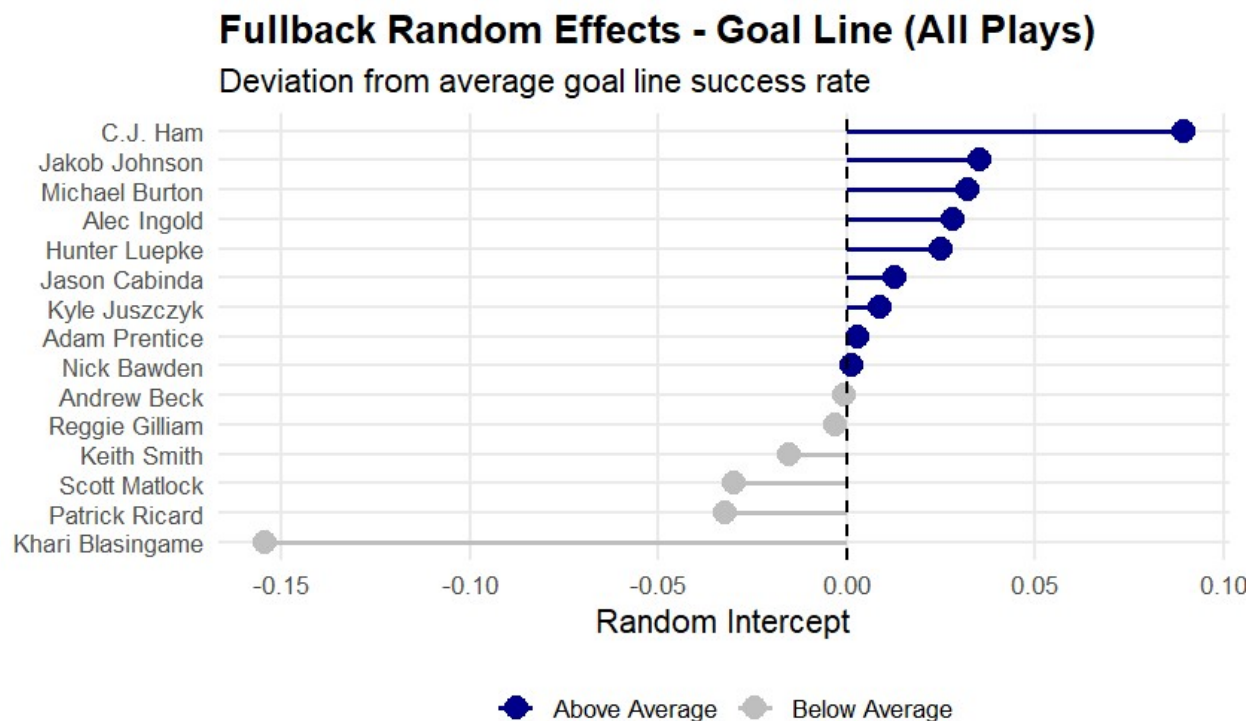
In addition to base models, a mixed effects model was used across all situations as well. Random intercepts and slopes were assigned to each fullback, so that the presence of a fullback could vary across all the situations. These models followed the same trend as the base models, being linear regression for non binary outcomes and logistic regression for binary outcomes. This allows for a better evaluation of fullbacks and how they can effect the three situations explored in this study. This model is represented by:

$$\log\left(\frac{p_{ij}}{1-p_{ij}}\right) = \beta_0 + \beta_1(has_fullback_{ij}) + \beta_2(down_{ij}) + \beta_3(ydstogo_{ij}) + \beta_4(yardline_100_{ij}) \dots + \beta_{11}(ypass_{ij})$$

The base down situation for the Mixed Effects model is represented by:

$$EPA_{ij} = \beta_0 + \beta_1(has_fullback_{ij}) + \beta_2(wpa_{ij}) + \beta_3(first_down_{ij}) + \beta_4(first_down_rush_{ij}) \dots + \beta_{11}(defense_coverage_type_{ij})$$

Also it allows you to see which fullbacks are above and below average in certain situations. Below you will see goal line all situations produced by the mixed effects model, and this plot will allow you to see, out of the fifteen fullbacks used in this study, who is below and above.



C.J. Ham and Kyle Juszczyk are profound fullbacks in the league and it is no surprise that they are above average. Players like Patrick Ricard raised some questions for me though, since he is a six time pro bowler. The conclusion I came up with as to why he is below average is that his game may be below average in this situation, but he makes his money in base down and short yardage situations. Also this is all plays as well, so he could either bring value in passing plays at the goal line or run plays as well, but this accounts for all plays.

Results

Before adding features in the model, Pearson correlation was used to see which outcome variables would go in my model. There was a lot of trouble at first with the p-values being not significant, so I used the Pearson correlation to try out new features to add to the model. For the short yardage situations, the strongest features were first down rush, with a score of .886, and first down with a score of .884. These were the features that resulted in a first down and if it were a rush play. Other features that were strongly correlated are wpa with a score of .664, drive first downs with a score of .418, and drive play count with a score of .378. These are the strongest correlations related to short yardage and ultimately resulted in the model results being significant. For goal line situations the strongest correlations were first down (.955), epa(.69), first down rush(.605). This shows that the plays that resulted in a touchdown are strongly correlated to these features that were used in the correlation model. Some honorable mention features in this were wpa and yardline_100. In base down situations the features were tested on whether or not they were strongly correlated with epa. The features include wpa(.775), first down rush(.285), first down(.645), drive first downs(.233). These were the strongest correlated features related to epa, and made the model significant in only rushing plays, which will be later shown. Before using the linear and logistic models I wanted to have a baseline of what to expect and see if the findings of a simple model would follow when the complex models were put to the test. Below you will find the results of the baseline model.

situation_type	plays_0	plays_1	avg_epa_0	avg_epa_1	success_rate_0	success_rate_1	avg.
base_down	84806	11299	0.001	0.011	0.422	0.416	
goal_line	6019	850	0.043	0.035	0.500	0.513	
short_yardage	7542	984	0.122	0.119	0.651	0.669	

These results show the success rate by having a fullback in across three situations. As you can see, goal line and short yardage success rates are higher with a fullback, and base down results have a minimal difference. As we dive into the complex models we will see how this holds up.

Base Model Short Yardage Results

Short Yardage - All Plays

Predictor	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-11.252	1.163	-9.676	0.000
has_fullback	0.584	0.223	2.621	0.009

Short yardage situations were modeled by logistic regression, and again the short_yardage_success variable was the response. A play is a success if you gained one yard on

third or fourth down. Three models were estimated, all short yardage, short yardage rushing plays, and short yardage passing plays. In the all short yardage model, the estimation for has_fullback was .584 with a p-value of .0088. This shows that having a fullback in all short yardage situations raises the likelihood of gaining one yard in these situations. This is a statistically significant model as well. For the rushing model, the estimation was .586 with a p-value of .046. This shows that having a fullback in rushing situations in short yardage raises your chances of gaining a yard. This is also a statistically significant model. For short yardage passing, the model estimated it out to be .849 with a p-value of .047. This shows that even in passing situations the presence of a fullback increases your chances of gaining that one yard. Overall, for short yardage situations, having a fullback is prominent for performance.

Base Goal Line Model Results

Goal Line - All Plays				
Predictor	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.040	0.008	4.984	0.000
has_fullback	-0.007	0.005	-1.568	0.117

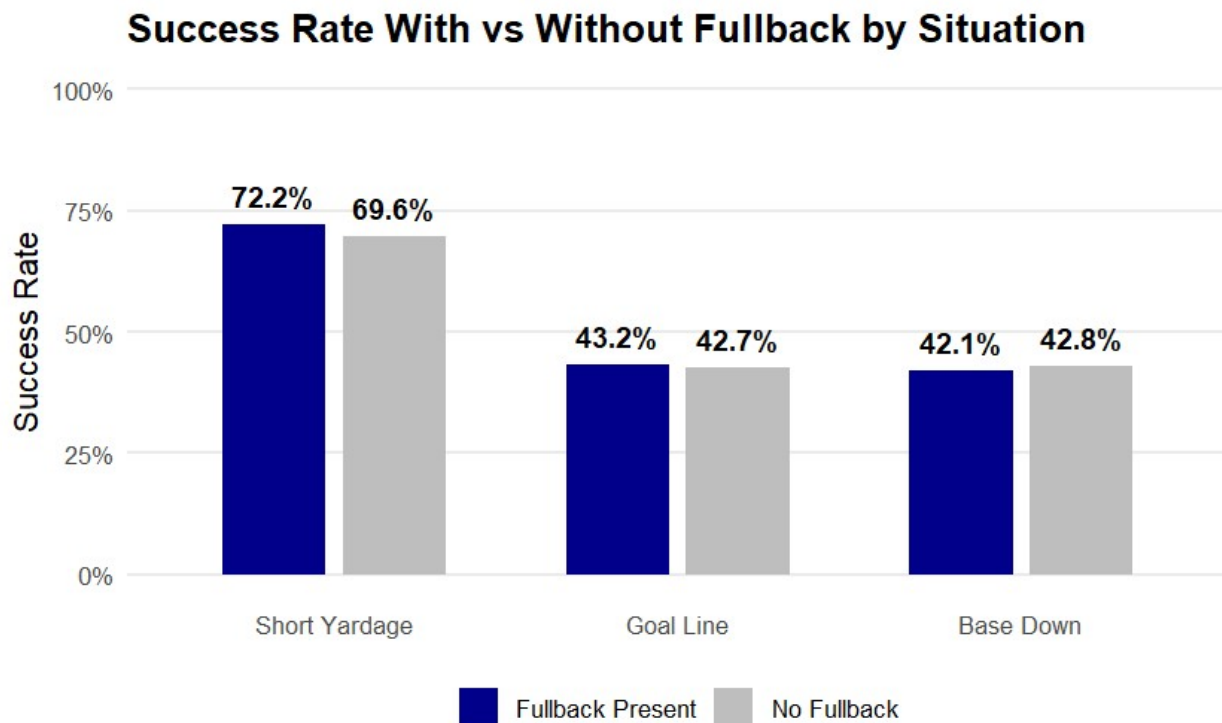
In goal line situations, the model used was logistic based regression as well, and success is defined as scoring a touchdown. Like the short yardage model, there were three models conducted as well, rushing, passing, and an overall model. In the overall model, the estimation for has_fullback was -.0074, with a p-value of .117. In the rushing model the estimation came out to be -.0061 with a p value of .338. Finally with the passing model the estimation was -.0086, with a p-value of .126. None of these models were statistically significant, so for all the models it shows that having a fullback in goal line situations does not affect the probability of scoring a touchdown.

Base Down Base Model Results

Base Down - All Plays				
Predictor	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0	0	-3.624000e+00	0.000
has_fullback	0	0	-1.101350e+02	0.000

For the base down models' linear regression was used to predict epa. Since this was a continuous variable linear regression was appropriate to use. Three models were estimated again, rushing,

passing, and overall model for base down. In the overall model, `has_fullback` was .0195, with a p-value of .091. This is a positive result and shows that having a fullback in the game in base downs overall will raise your chances of getting a better epa. This is also on the 10% margin, so it is only marginally significant. The base down rushing model estimated .0089, and the p-value was .156. This shows that having a fullback does not elevate epa. This is also not a statistically significant model, which indicates again that it is not prominent for a fullback to be in the game in for base downs in rushing situations. The base down passing model produced a score remarkably close to zero, just like the p value as well. This shows that having a fullback on the field in passing base downs does not influence the play. Also to answer the question that was posed earlier, the success rates held true using these models. You will be able to explore this below:



Mixed Effects Models Short Yardage Results

Short Yardage - All Plays

Predictor	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.921	1.007	-0.915	0.360
<code>has_fullback</code>	-0.090	0.213	-0.423	0.672

To achieve variation across fullbacks, the mixed effects model was put into place. This allowed the model to seek the differences between each fullback. Compared to the base model, the mixed

effects model for short yardage posted a negative coefficient $-.090$ and was not statistically significant with a p value of 0.672 . The rushing and passing model followed the same trend as well with a coefficient of $-.165$ and $.636$ respectively, and p values of $.570$ and $.202$. The random effects showed near zero variance for most of the models, showing that it is minimal variation between fullbacks in short yardage situations. The findings in this model show that once you account for player variability, the positive effect for fullbacks in short yardage situations vanishes, which means the earlier results were swayed by team specific factors and not the fullback itself.

Mixed Effects Model Goal Line Results

Goal Line - All Plays

Predictor	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.906	0.665	-5.877	0.000
has_fullback	0.594	0.301	1.975	0.048

The mixed effects model for goal line situations produced contrasting results to the base models in the previous sections. The goal line all model, *has_fullback* was positive being $.594$ and had a statistically significant p value of $.048$, meaning having a fullback present in goal line situations increases the probability of scoring touchdowns. For the rushing model, adding effects did not produce statistically significant results at all, and had other issues as well. The passing model showed impressive results as well posting an effect of 1.243 , and a p value of $.015$, showing that fullbacks provide value in passing situations at the goal line. These results are more apparent through player variability, which we could not see in our base down models.

Mixed Effects Model Base Down Results

Base Down - Rush Plays

Predictor	Estimate	Std. Error	t value
(Intercept)	-0.125	0.087	-1.430
has_fullback	0.011	0.014	0.782

For the base down results the effects model showed that *has_fullback* does not produce noteworthy results. Having a fullback in the game did not have an impact on EPA across all models explored. The overall model posted a coefficient of $-.011$, while rushing and passing posted $.011$ and $-.010$. All the models were zero and had no statistically significant p-values. This shows that adding variability added minimal effect to base down situations and shows that

having a fullback in the game will not necessarily hurt or help you. So, it will be best to leave them out so coaches can have more variation with play calls in the base down situations.

Conclusion

Overall usage of the fullback varies between which situation and model you choose. The strongest models to suggest that a fullback is needed are the short yardage situations in the base down models. The strongest models that suggest you need a fullback for mixed effects are the goal line models. For the short yardage results, this elevates the probability of getting one yard on a play, and for the goal line it increases your chances of scoring a touchdown. For goal line situations, the fullbacks are not needed according to base models, they have a negative effect on scoring a touchdown in these situations. In base down situations for the base model, having a fullback on the field produces a small effect on epa in these situations. This holds true in the mixed effects models as well, the effect is zero and it neither hurts nor helps the situation so it is best to keep them off the field, so play callers can have their full arsenal instead of a select few plays for the fullbacks. The results for base down base model are not statistically significant, only the overall model is in the 10% margin, but the others have no effect. In general, if a team needs to gain a small amount of yardage, fullbacks are the way to go. Adding variability and seeing which fullback is in the game, having a fullback in goal line situations is the way to go. Especially in the pass, you can set up action plays to catch the heavy defense off guard. It really opens the play book down at the goal line since you are very condensed, so having a fullback allows the play caller to get creative and deceitful. These situations are highly intense situations where it can change a game completely. In these situations, fullbacks flourish and can swing momentum towards the team. Momentum is not something you can compute, but playing the game getting those situations and extending drives puts teams in the driver seat.